

AALOK MANDAL

PERSONAL
PORTFOLIO WEBSITE

- <https://akm164.github.io/My3DWebsite/>

PROFESSIONAL
SUMMARY

High-achieving University of Wollongong Computer Science graduate double-majoring in Software Engineering and AI & Big Data, relocating to Darwin to pursue postgraduate engineering studies at Charles Darwin University. Offers a strong technical foundation in full-stack web and mobile development, with hands-on experience designing intuitive, responsive UIs and building robust RESTful APIs.

Proficient in modern frontend ecosystems, continuous deployment pipelines (CI/CD), and backend system integration. A fast, analytical learner with excellent cross-functional communication skills, highly motivated to apply core software engineering and automation capabilities to the NAFI team while expanding expertise in spatial data frameworks.

EDUCATION

- Master of Engineering

[2026 – present]

Charles Darwin University, Casuarina, NT

Electrical And Electronics Engineering
- Bachelor of Computer Science

[2024 – 2026]

University of Wollongong, Wollongong, NSW

Double Major: Software Engineering & Artificial Intelligence & Big Data
- Diploma of Information Technology

[2023 – 2024]

UOW College, Wollongong, NSW

Equivalent to Year 1 Computer Science, credited toward SENG & AIBD majors

TECHNICAL
SKILLS

- Python
 - React
 - Next.js
 - WordPress & Wix
- JavaScript
 - AWS Cloud tools
 - HTML/CSS
 - Web & UI Design
- CI/CD pipelines & Docker
 - GitHub
 - RESTful API
 - SQL

SOFT SKILLS

- Leadership
 - Communication
 - Quality Assurance
- Teamwork
 - Time Management
 - Agile collaboration
- Detail Orientation
 - Adaptability
 - Problem-Solving

WORK HISTORY

- Software Developer Intern

[11/2025 – 05/2026]

Realta Logic, Sydney, NSW

 - Developed and refactored Python modules for essential application features in collaboration with a senior software engineer.
 - Participated in daily stand-ups, planning sessions, and retrospectives while working in Agile sprints with a multi-developer team.
 - Authored and maintained unit and integration tests using pytest, boosting code reliability and streamlining the CI/CD pipeline.
 - Conducted code reviews and managed feature branches in Git, resolving merge conflicts and ensuring adherence to coding standards.

Freelance Web UI Designer & Developer

[07/2025 – Present]

Self-Employed, Wollongong, NSW

- Collaborated directly with clients and business stakeholders to gather technical requirements, translating project goals into intuitive, user-centric wireframes and page flows.
- Built responsive, mobile-first layouts using custom front-end styling stacks (HTML5, CSS3, JavaScript), ensuring clean UI architecture and strict cross-browser compatibility.
- Performed custom theme editing, styling updates, code overrides, and API integrations within WordPress and Shopify ecosystems to extend core platform capabilities.
- Audited and optimized web assets for accessibility, SEO fundamentals, and rendering performance, ensuring fast load times and seamless, easy-to-maintain client handovers.

Head Chef

[07/2022 – 04/2025]

Ariels Cafe, Wollongong, NSW

- **Team Leadership:** Guided and motivated a diverse kitchen team, fostering a positive, respectful, and high-performance culture.
- **Clear Communication:** Coordinated effectively with FOH, management, and suppliers to ensure smooth service and accurate order flow.
- **Calm Under Pressure:** Maintained composure during peak hours, prioritizing tasks and supporting staff to deliver consistent quality.
- **Problem-Solving:** Resolved last-minute issues such as shortages, equipment faults, or staff gaps with quick, practical decisions.
- **Training & Mentoring:** Developed junior chefs through hands-on coaching, constructive feedback, and structured skill progression.
- **Organization & Time Management:** Managed prep schedules, service timing, and workflow planning to keep the kitchen efficient and well-coordinated.

KNOWLEDGE & PROJECTS

- **Web Development:** Developed and maintained the commercial website for a local family business. Utilized HTML, CSS, and JavaScript, integrated CMS APIs, and optimized cross-browser layouts to deliver seamless, user-centric experiences.
- **Capstone Project** *University of Wollongong, 2026*
Skin Check Champions | Full-stack PWA for skin health, clinic discovery, and community engagement
 - Built a mobile-first progressive web app for skin health tracking, clinic search, bookings, and user engagement across Australia.
 - Developed responsive React and Next.js interfaces with reusable component architecture for public, authenticated, and admin workflows.
 - Implemented API-driven backend features with TypeScript, Drizzle ORM, PostgreSQL, and schema validation for user profiles, skin checks, referrals, and bookings.
 - Integrated authentication, transactional email, monitoring, and external service workflows to support a reliable production-ready platform.
 - Supported geospatial clinic discovery and data syncing, including Health Direct/FHIR-based clinic information and map-enabled search experiences.
 - Worked with modern delivery practices including testing, linting, type checking, and deployment automation.
- **DevOps & Cloud Solutions:** Partnered with academic faculty on cloud architecture initiatives to automate software build and deployment workflows. Utilized GitHub Actions to build and test robust CI/CD pipelines, gaining practical exposure to the foundational containerization and orchestration concepts required for GitLab and Docker environments. Provisioned cloud-native services across AWS environments to manage scalable web applications.

References will be provided upon request.